Stage 2: During braces

Bone graft for a cleft in the gum: the three stages

This model shows the same jaw about 2 to 3 years after the bone graft. The child is about 12 to 13 years old. The eye tooth has come down through the new bone. Fixed braces line up the teeth.

PLACEHOLDER: picture of the During braces model

Whole jaw: braces on, with the false tooth

Render with tools/sheet/render.js once the model is built.

Whole jaw: braces on, with the false tooth

PLACEHOLDER: picture of the During braces model

Braces taken off: the gap where the small tooth never grew

Render with tools/sheet/render.js once the model is built.

Braces taken off: the gap where the small tooth never grew

PLACEHOLDER: picture of the During braces model

From below: the roof of the mouth

Render with tools/sheet/render.js once the model is built.

From below: the roof of the mouth

PLACEHOLDER: picture of the During braces model

Left side: the eye tooth in the row of teeth

Render with tools/sheet/render.js once the model is built.

Left side: the eye tooth in the row of teeth

What has changed

Bone now joins the two parts of the upper jaw where the gap was.[1,2]

The eye tooth came down through the graft into the row of teeth.[3,4]

Most eye teeth come through on their own. Some get stuck. Then a small operation uncovers the tooth, and the braces slowly pull it down.[4,5,6]

Some of the graft shrinks over time. Teeth that move into the graft help keep the bone.[7,8]

The braces

Fixed braces line up the teeth. Almost everyone who has this graft needs them.[1,2]

A false tooth on the brace wire fills the gap where the small tooth never grew. It holds the space until the braces come off.

The pictures show the jaw with and without the braces.

After the operation

In the first weeks, the gum heals over the graft. If the wound opens, you can lose some of the graft.[9,10]

An X-ray at about 3 to 6 months shows how well the new bone has taken.[11,12]

Your hip can be sore for a while after the surgeon takes the bone. In one study, children went back to school about 2 weeks after surgery.[13]

Sources **1.** Bergland O, Semb G, Abyholm FE. Elimination of the residual alveolar cleft by secondary bone grafting and subsequent orthodontic treatment. Cleft Palate J. 1986;23:175-205. PMID 3524905. pubmed.ncbi.nlm.nih.gov/3524905 **2.** Semb G. Alveolar bone grafting. Front Oral Biol. 2012;16:124-136. doi.org/10.1159/000337666 **3.** Rivers J, et al. A clinical index scale for secondary alveolar bone graft success based on canine eruption. Eur J Orthod. 2025;47:cjae065. doi.org/10.1093/ejo/cjae065 **4.** Vellone V, et al. Canine eruption after secondary alveolar bone graft in unilateral cleft lip and palate patients. J Craniofac Surg. 2017;28:1206-1210. doi.org/10.1097/SCS.0000000000003659 **5.** Hoang L, et al. Factors contributing to canine impaction in unilateral cleft lip and palate patients undergoing alveolar bone grafts. J Oral Maxillofac Surg. 2023;81:1286-1294. doi.org/10.1016/j.joms.2023.06.029 **6.** Parkin N, et al. Open versus closed surgical exposure of canine teeth that are displaced in the roof of the mouth. Cochrane Database Syst Rev. 2017;(8):CD006966. doi.org/10.1002/14651858.CD006966.pub3 **7.** Jahanbin A, et al. Long-term stability of alveolar bone graft in cleft lip and palate: a systematic review and meta-analysis. J Craniofac Surg. 2022;33:e194-e200. doi.org/10.1097/SCS.0000000000008254 **8.** Schultze-Mosgau S, et al. Bone resorption after secondary alveolar cleft bone grafts before and after canine eruption. J Oral Maxillofac Surg. 2003;61:1245-1248. doi.org/10.1016/s0278-2391(03)00722-5 **9.** Siddiqui S, et al. Autologous platelet concentrate during secondary alveolar bone grafting and wound dehiscence: a systematic review. J Craniomaxillofac Surg. 2024;52:895-905. doi.org/10.1016/j.jcms.2024.04.016 **10.** Kaura AS, et al. Optimal timing of alveolar cleft bone grafting. J Craniofac Surg. 2018;29:1551-1557. doi.org/10.1097/SCS.0000000000004680 **11.** Kindelan JD, et al. Radiographic assessment of secondary autogenous alveolar bone grafting. Cleft Palate Craniofac J. 1997;34:195-198. doi.org/10.1597/1545-1569(1997)034<0195:RAOSAA>2.3.CO;2 **12.** Witherow H, et al. A new scale to assess radiographic success of secondary alveolar bone grafts. Cleft Palate Craniofac J. 2002;39:255-260. doi.org/10.1597/1545-1569(2002)039<0255:ANSTAR>2.0.CO;2 **13.** Tache A, et al. Pain management at iliac donor sites after grafting of alveolar clefts. Int J Oral Maxillofac Surg. 2022;51:62-69. doi.org/10.1016/j.ijom.2021.05.004

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